# Timestamped Presenter Script for the Silent ITCT Reproduction Demonstration

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**Verified video duration: 10:14.17 | 1920 × 1080 | 30 fps | no audio stream**

This document supplies the complete live narration for ITCT_Reproduction_Demonstration.mp4. The timestamps follow the rendered scene boundaries. A natural speaking rate of approximately 110 to 130 words per minute leaves time for the bracketed visual cues and pauses. Do not add this script as an audio track; the submitted MP4 must remain silent.

## Scene 1  00:00.00 to 00:45.00  Project scope and evidence

**Visible elements**  Project title, assigned paper, contributor names and IDs, and the verified long-preset configuration.

**Word-for-word narration**  This project reproduces Opening the Black Box of Deep Neural Networks via Information by Ravid Shwartz-Ziv and Naftali Tishby. We ask what happens to the information stored by hidden layers while a network learns. Our evidence comes from the submitted NumPy notebook: twenty independently initialized networks, ten thousand training epochs, and a thirty-bin mutual-information estimator. We will separate direct measurements from theoretical interpretations. In particular, a decrease in binned information does not automatically prove an equal decrease in continuous mutual information. The video is silent so that we can explain the experiment live while the verified notebook outputs remain visible.

**Visual cues**  [READ THE TITLE] [POINT TO BOTH CONTRIBUTORS] [PAUSE ON 20 NETWORKS AND 10,000 EPOCHS] [NEXT SCENE]

**Timing note**  Scene duration is 45.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 2  00:45.00 to 01:30.00  Synthetic task and measurement

**Visible elements**  The icosahedron feature, logistic label calibration, and conditional label probabilities from the submitted notebook.

**Word-for-word narration**  The synthetic task contains twelve binary inputs, so the complete input universe has two to the twelfth power, or four thousand ninety-six patterns. The coordinates correspond to vertices of an icosahedron. A rotation-invariant feature groups the patterns into sixty-four feature classes. The notebook applies a logistic label rule with gain gamma equal to eighty-five point three. It chooses the threshold so that the positive-label probability is approximately zero point four nine seven and the available information I of X semicolon Y is approximately zero point nine nine zero bits. Training labels are sampled from this soft rule. Information estimates use the known population distribution over every input pattern.

**Visual cues**  [POINT TO 4,096 PATTERNS] [TRACE THE LOGISTIC CURVE] [INDICATE THE LABEL-PROBABILITY PANEL] [NEXT SCENE]

**Timing note**  Scene duration is 45.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 3  01:30.00 to 03:09.17  Main information-plane trajectory

**Visible elements**  An animated information plane with twenty network observations per layer, layer means as diamonds, the IB frontier, current epoch, phase, and numerical layer means.

**Word-for-word narration**  The information plane gives every layer two coordinates. The horizontal axis is I of X semicolon T, the information retained about the input after quantization. The vertical axis is I of T semicolon Y, the information retained about the target. Small points show the twenty individual networks, and diamonds show the ensemble mean. At early epochs, deeper layers have little label information. During fitting, the points move upward as the network becomes predictive. The operational gradient signal-to-noise transition occurs near epoch eighty-four. After that point, narrower layers move left while remaining near the top. Under this estimator, that means fewer input patterns receive distinct binned activation codes while target information remains high. At the final epoch, the output layer is near one point one five bits horizontally and zero point nine four three bits vertically. The widest layers remain close to the twelve-bit ceiling. Notice that the video advances only through the one hundred nineteen saved notebook frames. It does not interpolate unmeasured experimental states. The observed leftward movement supports binned compression in selected layers. It does not by itself establish continuous compression or a causal diffusion mechanism.

**Visual cues**  [POINT TO THE HORIZONTAL AXIS] [POINT TO THE VERTICAL AXIS] [WATCH THE DIAMONDS RISE] [HIGHLIGHT EPOCH 84] [OBSERVE LAYERS 4 TO 6 MOVE LEFT] [PAUSE AT THE FINAL FRAME] [NEXT SCENE]

**Timing note**  Scene duration is 99.17 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 4  03:09.17 to 04:04.17  SGD drift and diffusion diagnostic

**Visible elements**  The notebook's gradient mean, gradient standard deviation, layerwise SNR, cross-entropy, information, and weight-norm curves.

**Word-for-word narration**  The paper interprets training as an early drift phase followed by a noise-dominated phase. The notebook measures this with a gradient signal-to-noise ratio: the norm of the mean minibatch gradient divided by the norm of its fluctuations. The initial geometric mean is about three point nine seven. The first post-peak crossing below one occurs near epoch eighty-four, earlier than the paper's approximate epoch three hundred fifty. About ninety-nine percent of the ten-thousand-epoch schedule lies after this crossing. Weight norms continue to increase, so the information decrease cannot be explained by simple weight shrinkage. The cautious conclusion is that the gradient statistic and binned information change together. This diagnostic does not prove Brownian diffusion or establish that gradient noise causes compression.

**Visual cues**  [HIGHLIGHT THE SNR PANEL] [POINT TO THE CROSSING BELOW ONE] [COMPARE EPOCH 84 WITH THE PAPER] [POINT TO INCREASING WEIGHT NORMS] [NEXT SCENE]

**Timing note**  Scene duration is 55.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 5  04:04.17 to 04:59.17  Information Bottleneck frontier

**Visible elements**  The computed Information Bottleneck frontier and the six converged layer endpoints.

**Word-for-word narration**  The Information Bottleneck objective balances input compression against target relevance. The mean vertical gap between the converged layers and the computed frontier is about zero point zero one zero bit, with a maximum near zero point zero four four. Vertical proximity alone is incomplete. Layers one through three retain roughly nine point six six to ten point five five excess input bits at comparable target information. The output layer retains only about zero point two excess bit. Therefore the deeper representations are economical in both coordinates, while wide layers remain highly redundant even when their vertical gap appears small. The reproduction gives the paper credit for the frontier comparison, but it avoids calling every vertically close point globally optimal.

**Visual cues**  [TRACE THE BLUE FRONTIER] [POINT TO THE WIDE LAYERS ON THE RIGHT] [COMPARE VERTICAL GAP WITH HORIZONTAL EXCESS] [NEXT SCENE]

**Timing note**  Scene duration is 55.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 6  04:59.17 to 05:54.17  Network-depth comparison

**Visible elements**  Depth-dependent information trajectories and the epoch counts required to reach fixed fractions of available label information.

**Word-for-word narration**  This experiment asks whether additional hidden layers accelerate representation learning. At the eighty-percent threshold, one hidden layer needs about one thousand five hundred forty-eight epochs. Two layers need six hundred twelve, and three layers need five hundred forty-two. The best observed reduction is therefore about two point nine times from one to three layers. Four, five, and six hidden layers require about six hundred nine, six hundred nine, and six hundred eighty-five epochs. The trend is not monotone after three layers. These values support a depth advantage over the shallow baseline, but not the unrestricted claim that every added layer trains faster. Epoch count also differs from wall-clock time and computation per epoch.

**Visual cues**  [COMPARE THE SIX TRAJECTORIES] [POINT TO 1,548 AND 542] [SHOW THE NON-MONOTONE TAIL] [NEXT SCENE]

**Timing note**  Scene duration is 55.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 7  05:54.17 to 06:49.17  Training-set-size comparison

**Visible elements**  Final output-layer I(T;Y) across training fractions from three to eighty-five percent.

**Word-for-word narration**  The training-set experiment evaluates every network on the complete four-thousand-ninety-six-pattern distribution while changing only how much of that universe the optimizer sees. The final output-layer label information rises from about zero point two three four bit at three percent coverage to zero point nine four three bit at eighty-five percent. Intermediate values increase through zero point two seven seven, zero point four zero one, zero point five eight one, zero point six eight five, zero point eight one one, and zero point eight seven six bits. This is population-level retained relevance, not training accuracy. More observed patterns improve recovery of the rule, but the experiment does not prove that leftward compression causes generalization.

**Visual cues**  [POINT TO THE THREE-PERCENT ENDPOINT] [MOVE ALONG THE FRACTION SEQUENCE] [PAUSE AT 0.943 BITS] [NEXT SCENE]

**Timing note**  Scene duration is 55.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 8  06:49.17 to 07:44.17  Committee-machine generality test

**Visible elements**  Information trajectories and gradient diagnostics for the non-symmetric committee-machine task.

**Word-for-word narration**  The generality test changes both the target rule and the student architecture. The teacher is a majority vote of five random perceptrons with a stochastic label rule calibrated to the same zero point nine nine zero-bit ceiling. The student network is twelve-eight-five-three-two and trains on eighty-five percent of the input patterns. Its gradient-SNR transition occurs near epoch seventy-five. The layers again gain target information before moving left under the quantized estimator. Similar behavior across the icosahedron and committee tasks provides a useful robustness check. It remains evidence from two small synthetic problems, not proof that every network or real-world dataset follows the same dynamics.

**Visual cues**  [IDENTIFY THE DIFFERENT ARCHITECTURE] [POINT TO THE TRANSITION NEAR 75] [COMPARE THE PATH SHAPE WITH THE MAIN TASK] [NEXT SCENE]

**Timing note**  Scene duration is 55.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 9  07:44.17 to 08:34.17  Tanh and ReLU comparison

**Visible elements**  Peak-to-final binned input-information curves for tanh and ReLU networks.

**Word-for-word narration**  The activation control asks whether the measured information decrease depends on saturating tanh units. Under the implemented rule, four of six tanh layers and five of six ReLU layers satisfy the practical drop criterion. That numerical result needs an important qualification. ReLU activations are unbounded, and the notebook recomputes the bin interval from zero to the current maximum at each snapshot. If a few activations grow, the bins become wider for most examples and the estimated entropy can fall without an equivalent loss of continuous distinguishability. A fixed-range quantizer was not run, so the ReLU result is evidence of estimator behavior, not a conclusive causal compression result.

**Visual cues**  [COMPARE TANH AND RELU CURVES] [POINT TO THE MOVING ZERO-TO-MAX RANGE LIMITATION] [NEXT SCENE]

**Timing note**  Scene duration is 50.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 10  08:34.17 to 09:24.17  Bin-count sensitivity

**Visible elements**  The same experiment evaluated with ten, thirty, and one hundred bins.

**Word-for-word narration**  Changing the number of bins changes both the absolute mutual-information estimates and the count of layers that meet the practical compression rule. Five layers qualify at ten bins, four at thirty bins, and three at one hundred bins. Layers four through six decrease under every setting, which gives selected qualitative robustness. The magnitude still changes materially. At one hundred bins, the two widest layers reach the twelve-bit input ceiling because the finer quantizer separates more activation vectors. We should therefore compare the direction and structure of the paths while reporting the estimator explicitly. The experiment rejects any claim that the numerical information values are independent of quantization.

**Visual cues**  [COMPARE THE THREE PANELS] [POINT TO THE TWELVE-BIT CEILING] [STATE FIVE, FOUR, AND THREE LAYERS] [NEXT SCENE]

**Timing note**  Scene duration is 50.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.

## Scene 11  09:24.17 to 10:14.17  Findings, limitations, and conclusion

**Visible elements**  A claim-level synthesis separating supported, operational, qualified, partial, and estimator-dependent conclusions.

**Word-for-word narration**  The strongest reproduced result is the layer-dependent transition from fitting toward lower binned input information. The gradient-SNR crossing gives a reproducible operational boundary, but it does not prove physical diffusion. The Information Bottleneck comparison is strongest for deep layers when both vertical gap and horizontal excess are considered. Depth provides a large improvement over one hidden layer, then becomes non-monotone. ReLU and bin-count experiments show that the quantizer is part of the measured phenomenon. Our conclusion is therefore positive but limited: the information plane is a useful descriptive framework for these controlled experiments, while continuous compression, causal diffusion, global IB optimality, and universal depth acceleration remain open. We will use the website and notebook to answer questions about any point shown here.

**Visual cues**  [READ EACH EVIDENCE CATEGORY] [PAUSE ON ESTIMATOR-DEPENDENT] [INVITE QUESTIONS]

**Timing note**  Scene duration is 50.00 seconds. Use the remaining time for the indicated pauses and pointing cues before the next scene begins.